SkillShift

An AI career news portal that traces tech news to real TPM hiring requirements

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Live app: skillshift-xi.vercel.app | Code: github.com/deokhwajeong/skillshift

> ***The idea in one line:*** *news tells you what changed, job postings tell you what is hired for. SkillShift connects the two, and lets you click through to the evidence behind every insight.*

1. Problem

Tech job seekers read a constant stream of news but cannot tell which shifts matter for hiring. News portals summarize what happened without saying what it means for a career. Resume tools match keywords to postings without explaining why a skill matters now. Generic chatbot advice ("learn AI evaluation") comes with no evidence the user can check.

2. Target User

Graduate students and early-career professionals preparing for Technical Program Manager (TPM) internships and full-time roles, who must decide what to learn, what to highlight on a resume, and what to prepare for interviews, with limited time.

3. Solution

SkillShift reads curated tech news, extracts each concrete *shift* (a change in technology, practice, regulation, or market), verifies it against the article, and checks it against what real TPM postings ask for. Every skill lands in a **Signal Matrix**: **Rising** (in the news and in postings), **Emerging** (in the news, not yet in postings: an early-mover chance), **Established** (in postings, not in the news: table stakes), or **Fading**. **News is the starting point; job postings are the evidence that grounds it.**

4. Key Features

Built in this prototype

- **Signal Matrix:** 20 TPM skills placed by news signal vs job-posting signal; the quadrant is a fixed rule in a config file, not the model's opinion
- **Today's Brief:** What changed / Why it matters for TPMs / What to do this week, every sentence cited; **Listen** reads it aloud
- **News-to-Skill traceability:** click a skill to see how its score was computed, word-for-word article quotes, and the exact posting lines, with AI interpretation labeled separately
- **Editor Queue (human review):** weak, low-confidence, or conflicting findings wait for a person; approving one re-scores the matrix
- **Run live + How it works:** one fresh article streamed through the agents; every stage shows real input and output, cost, and a link to its code

Future: more roles (PM, SWE, Data Science), live feeds, resume gap analysis, weekly digest

5. AI / Technical Approach

A seven-stage pipeline on the Claude API (Claude Sonnet 5.5), built with Next.js, TypeScript, and Zod, deployed on Vercel. **LLMs read; code decides.**

- **Filter** (LLM): relevance and duplicates; code applies the threshold
- **Shift Extraction** (LLM): each shift with a short quote, skills, confidence
- **Verification** (LLM + code): is the quote really in the article, is it supported, does it conflict with other sources
- **JD Analyzer** (LLM + code): required vs preferred skills per posting
- **Signal Scorer** (code): recency- and source-weighted scores, quadrants
- **Brief Writer** (LLM): the cited briefing

Every LLM output is schema-validated with one retry; anything that fails or is uncertain goes to the Editor Queue, never straight to the page.

Data: 14 recent articles (Aug to Oct 2026; company blogs, trade press, independent blogs) and 16 real TPM postings (Anthropic, Stripe, Airbnb, Waymo, and others). **Real run:** 11 articles kept, 49 shifts, 38 published, 11 held for review; 53 LLM calls, about $0.75. Results are precomputed for demo reliability.

6. Demo Flow

- Persona: a CMU student preparing for TPM recruiting
- Today: the matrix and the cited brief, read aloud
- Click a skill and trace it to source quotes and postings
- Run live: a fresh article (InfoQ, Oct 24): 2 shifts published, 3 held
- Try a weak input: why a vague article never gets published
- How it works: stage counts, real inputs and outputs, View code

7. Limitations

- Small hand-curated dataset: signals describe this set, not the whole market, and are not predictions
- LLMs can misread or over-connect; quote checks, verification, and human review reduce this but do not eliminate it
- Accuracy on a small labeled set: 96% precision, 100% recall (11 articles); 98% / 100% (5 postings). I labeled by reviewing the agents' output, so these are likely optimistic
- Limited to TPM roles and English-language sources

8. Future Direction

As a semester team project, **each member owns a role vertical** (PM, SWE, Data Science), so the product grows with the team's own career interests. Next: larger and live data, trend validation over time, personalized learning paths, and testing with CMU students during recruiting season. **Business model:** licensing to university career centers and bootcamps, plus a free tier for individual job seekers.